

PRODUCT REQUIREMENTS DOCUMENT (PRD)

Project Guardian: AI-Based Fake Identity & Document Screening System

PS ID: 26188 Client: Ministry of Home Affairs – Sashastra Seema Bal (SSB)

Doc Version: 1.0.0 (Production Blueprint) Theme: Blockchain & Cybersecurity

1. Document Control & Executive Metadata

Document Title	Product Requirements Document (PRD)	Project Codename	Project Guardian (SIH PS 26188)
Target Ministry	Ministry of Home Affairs (MHA)	Department / Force	Sashastra Seema Bal (SSB), Police II Division
Problem Category	Software	Thematic Area	Blockchain & Cybersecurity
Classification	Confidential — Prototype Specification	Status	Approved for Implementation

2. Executive Summary & Problem Definition

2.1 Background & Operating Context

The **Sashastra Seema Bal (SSB)** oversees border security across India's open, visa-free borders with Nepal (1,751 km) and Bhutan (699 km). Border outposts and Integrated Check Posts (ICPs) handle high pedestrian and vehicular traffic daily. Travelers present a heterogeneous mix of documents: Indian and Nepalese/Bhutanese citizens carry domestic identity cards (Voter ID/EPIC, Aadhaar, Driving Licenses, Transit Permits), while third-country foreigners carry international Passports and Visas.

2.2 Problem Statement & Core Operational Vulnerabilities

Current border verification relies on visual human inspection and isolated database lookups. This operational model suffers from critical vulnerabilities:

- High Passenger Bottlenecks:** Manual checks take 2–5 minutes per person, leading to severe congestion.
- Undetected Physical & Digital Tampering:** High-quality color photo replacements, edited dates of birth, and counterfeit visa ink stamps cannot be caught by visual inspection alone.
- Identity Impersonation:** Lookalike individuals travel on genuine documents of other persons.
- Multiple Identities by the Same Person:** Trans-border criminals and smugglers create multiple identity cards across different jurisdictions under different names.
- Lack of an Immutable Audit Trail:** Border logs on traditional spreadsheets or isolated servers are vulnerable to unauthorized modification or repudiation.

2.3 Solution Vision

Develop **Project Guardian**, an integrated, AI-driven, and blockchain-audited border screening platform that captures document images and live face feeds, executes automated perspective correction, extracts dual-zone OCR data, validates standard checksums (ICAO 9303, Aadhaar Verhoeff), detects multi-signal digital forgeries (ELA + ViT + EXIF), verifies biometrics (1:1 match and 1:N FAISS duplicate search), and outputs an explainable risk score (0–100) with immutable ledger commit within **under 2.5 seconds**.

3. Key Performance Indicators (KPIs) & Success Metrics

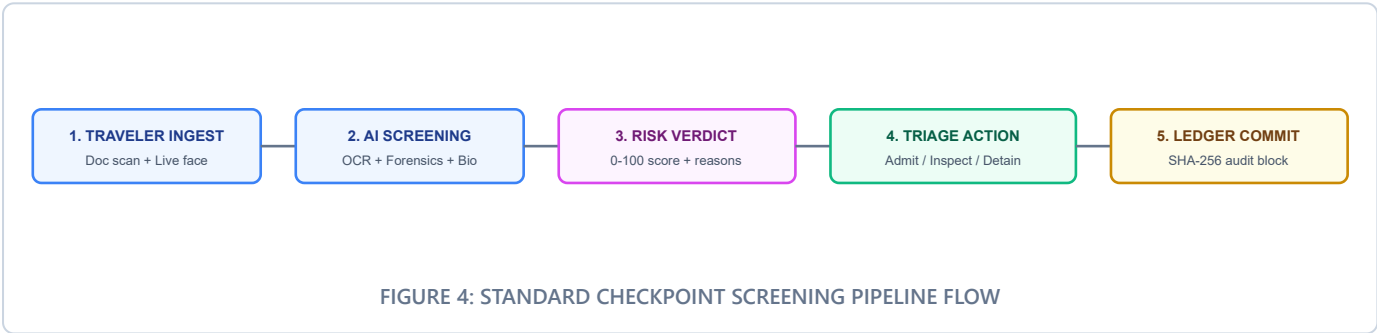
Metric / KPI	Target Threshold	Operational Significance
Total Verification Latency	≤ 2.50 seconds	End-to-end processing per traveler to prevent checkpoint bottlenecks.
MRZ Checksum Accuracy	100.0%	Zero false calculations on ICAO 9303 TD3 standard 7-3-1 check digits.
OCR Field Accuracy	≥ 98.5%	Correct character extraction on high-resolution deskewed identity documents.
Tamper Detection F1-Score	≥ 92.0%	Balanced precision and recall across photo replacement, text alteration, and Canva/PS edits.
Biometric False Match (FMR)	≤ 0.1%	Prevents impostors from passing face verification checks.
1:N Duplicate Search Time	≤ 50 ms	FAISS index search speed across 10,000 enrolled border profiles.
Audit Trail Integrity	100.0%	Zero repudiation; tamper-evident SHA-256 block chain commit on every screening event.

4. Target User Personas & Detailed User Journeys

4.1 User Personas

PERSONA	ROLE & ENVIRONMENT	PRIMARY OPERATIONAL NEEDS
SSB Ground Officer	Checkpoint booth duty at border ICP (e.g. Raxaul, Jogbani). High passenger volume.	Sub-second result, clear visual cues (Green / Amber / Red banner), automated camera capture, minimal manual typing.
Supervisory / Intelligence Officer	Secondary inspection room. Handles flagged/detained travelers.	Deep evidence drill-down: side-by-side ELA compression heatmap toggle, VIZ-to-MRZ discrepancy viewer, FAISS biometric duplicate alerts.
System Auditor / Cyber Cell	Command HQ / State Police II IT Division.	Tamper-evident audit ledger inspection, verification of block hashes, screening receipts, officer badge attribution.

4.2 Primary Screening User Journey



5. System Scope & Document Taxonomy

SCOPE CATEGORY	SUPPORTED DOCUMENT TYPES	IN SCOPE / OUT OF SCOPE
International Travel	ICAO Doc 9303 TD3 Passports, Indian Sticker Visas, e-Visa Grant Letters.	IN SCOPE Full MRZ, Checksum, ELA, ViT.
National Identification	Election Commission Voter ID (EPIC), UIDAI Aadhaar Card.	IN SCOPE Verhoeff check, regex, tamper checks.
Driving Licenses	Sarathi State Transport Smart Cards / Laminated DLs.	IN SCOPE State regex, validity bounds.
Border Permits	SSB Emergency Transit Slips, Embassy Identity Slips.	IN SCOPE Stamp and date validation.
Physical Contactless Chips	Physical ISO 14443 NFC chip hardware reading via browser.	OUT OF SCOPE Simulated LDS data verification.
Classified Police Networks	Direct live network write access to CCTNS / Interpol SLTD.	OUT OF SCOPE High-fidelity local SQLite mock.

6. Detailed Functional Requirements (FR-01 to FR-08)

FR-01	Automated Document Ingestion & Computer Vision Preprocessing	Core CV
Description: The system must accept camera/scanner uploads (JPEG, PNG, WEBP), identify quadrilateral document borders, and deskew the image to a standardized top-down orientation. FR-01.1: Automatically rescale high-resolution images to target height (1000px) and apply Gaussian blur with Canny edge detection. FR-01.2: Locate the largest convex contour using <code>cv2.approxPolyDP</code> and compute a 4-point perspective transform (<code>cv2.warpPerspective</code>). FR-01.3: Apply Contrast Limited Adaptive Histogram Equalization (CLAHE) on the LAB L-channel to eliminate glare and shadows.		
FR-02	Dual-Zone Optical Character Recognition (OCR) Engine	Core OCR
Description: The system must extract text across both structured Machine Readable Zones (MRZ) and unstructured Visual Inspection Zones (VIZ). FR-02.1: Parse 2-line TD3 MRZs via <code>PassportEye</code> and extract Name, Passport Number, Nationality, DOB, Expiry, Gender, and Check Digits. FR-02.2: Apply scoped positional letter-to-digit auto-correction strictly to numeric positions (DOB: 13-19, Expiry: 21-27, Check digits: 9, 19, 27, 43): <code>O / D → 0</code>, <code>I / L → 1</code>, <code>Z → 2</code>, <code>A → 4</code>, <code>S → 5</code>, <code>B → 8</code>. FR-02.3: Fall back to <code>PaddleOCR</code> / <code>Tesseract</code> for VIZ text extraction and regex pattern matching on Voter IDs and Driving Licenses. FR-02.4: Flag character-level extraction plausibility with confidence ratings (High / Low).		

FR-03

Deterministic Standards & Rule-Based Validation Engine

Rule Engine

Description: Verify that extracted document data complies with international and national mathematical validation standards.

- **FR-03.1:** Compute ICAO Doc 9303 7-3-1 modulo 10 checksums on Passport Number, Date of Birth, Date of Expiry, and Composite Line 2.
- **FR-03.2:** Implement the **Aadhaar Verhoeff algorithm** over the dihedral group D_5 to validate 12-digit Indian national identity numbers.
- **FR-03.3:** Execute **VIZ-to-MRZ Cross-Field Consistency Checks**: Compare printed visual name and dates against MRZ decoded values.
- **FR-03.4:** Validate date plausibility: Expiration date > Current date; Date of Birth not in future and < 120 years ago; Issue date < Expiration date.

FR-04

Multi-Signal Digital Forgery & Tampering Detection Engine

Core AI Innovation

Description: Detect pixel alterations, photo swaps, digital modifications, and re-compression anomalies across four distinct attack vectors.

- **FR-04.1 (ELA Analysis):** Re-save input at JPEG Quality 90, compute absolute pixel variance, amplify contrast, save a PNG difference heatmap, and return a normalized 0–100 ELA score.
- **FR-04.2 (AI ViT Classifier):** Blend the original image with ELA (alpha=0.3) and execute classification using pretrained Vision Transformer `zodumair/document-forgery-detector` to output a deepfake/forgery probability.
- **FR-04.3 (EXIF Forensics):** Parse EXIF tags for graphic editing tools (Photoshop, GIMP, Canva, Snapseed) and flag stripped/missing metadata.
- **FR-04.4 (Stamp & Seal Forensics):** Detect overland immigration stamps and evaluate color histogram consistency and contour edge discontinuities.

FR-05

Biometric Face Verification (1:1 Matching)

Biometrics

Description: Verify that the individual standing at the checkpoint matches the photographic portrait printed on the document.

- **FR-05.1:** Automatically crop the document portrait photo using facial boundary coordinates or OpenCV cascade detection.
- **FR-05.2:** Capture live webcam video frames from the browser via HTML5 `MediaDevices.getUserMedia`.
- **FR-05.3:** Compute 512-dimensional facial embeddings using deep ArcFace / MobileFaceNet networks and evaluate cosine similarity.
- **FR-05.4:** Apply calibrated operational decision thresholds: ≥ 0.65 (Verified Match), 0.50–0.64 (Uncertain / Manual Check), < 0.50 (Mismatch / Impersonation Alert).

FR-06

Biometric Duplicate Identity Detection (1:N FAISS Vector Search)

Anti-Alias AI

Description: Detect trans-border individuals attempting to cross under multiple identities, aliases, or forged names.

- **FR-06.1:** Maintain an in-memory **FAISS** (Facebook AI Similarity Search) index storing embeddings of previously screened travelers.
- **FR-06.2:** On each live face capture, perform an inner-product nearest-neighbor search against the index within $\leq 50\text{ms}$.
- **FR-06.3:** If a matching biometric face (> 0.70 similarity) is discovered under a *different name, nationality, or passport number*, immediately trigger a **"CRITICAL: Duplicate Identity / Alias Detected"** alert.

FR-07

Explainable Evidence Fusion & Risk-Scoring Engine

XAI Decision

Description: Consolidate multi-modal signals into an auditable 0–100 risk score accompanied by transparent reason codes.

- **FR-07.1:** Apply dynamic penalty weights: Blacklist (+50), Biometric Mismatch (+30), Checksum Failure (+25), Document Expired (+20), EXIF Edit Tool (+15), VIZ-MRZ Mismatch (+15).
- **FR-07.2:** Categorize decision into 3 triage bands: **LOW (< 30)**: Auto-Admit; **MEDIUM (30–60)**: Secondary Inspection; **HIGH (> 60)**: Immediate Detention.
- **FR-07.3:** Output an array of plain-English diagnostic reason codes for immediate officer understanding.

FR-08

Blockchain & Cryptographic Audit Trail Layer

Blockchain & Security

Description: Ensure non-repudiation and immutable logging of every border screening event in compliance with the SIH theme.

- **FR-08.1 (Zero-PII On-Chain):** Store document images and raw biometrics off-chain. Commit only SHA-256 document hashes, risk scores, model versions, and officer badge IDs.
- **FR-08.2 (Cryptographic Block Hash):** Link each event into an append-only hash chain: `Block_Hash = SHA-256(Index + Timestamp + Prev_Hash + Doc_Hash + RiskScore + OfficerID)`.
- **FR-08.3 (Ledger Integrity API):** Expose `GET /api/ledger/verify` to validate that the cryptographic hash chain is uncorrupted and unbroken.

7. Non-Functional Requirements (NFRs)

NFR AREA	TARGET SPECIFICATION	IMPLEMENTATION DETAIL
Latency Budget	≤ 2.50s End-to-End	Preprocessing: ≤ 100ms; OCR/MRZ: ≤ 600ms; ELA + ViT: ≤ 800ms; Biometrics: ≤ 400ms; Fusion + Ledger: ≤ 50ms.
Security & Privacy	DPDP Act 2023 Compliant	AES-256 encryption at rest; TLS 1.3 in transit; live face frames processed in memory and immediately discarded.
Offline Resilience	100% Local Autonomous	Containerized via Docker; local SQLite registry ensures operation during border outpost telecom outages.
UI/UX Ergonomics	High-Visibility Dashboard	Executive light theme with high-contrast navy headers; responsive for desktop workstations and ruggedized border tablets.
System Scalability	100+ concurrent streams	Asynchronous FastAPI endpoints powered by Uvicorn ASGI workers.

8. Technical Architecture & Database Schemas

8.1 Complete Database Schema (SQLite Off-Chain Store)

```
-- 1. Document Registry Table
CREATE TABLE IF NOT EXISTS documents (
  document_id TEXT PRIMARY KEY,
  document_number_hash TEXT NOT NULL,
  issuing_country TEXT NOT NULL,
  document_type TEXT NOT NULL,
  issue_date TEXT,
  expiry_date TEXT,
  status TEXT NOT NULL CHECK(status IN ('VALID', 'EXPIRED', 'REVOKED', 'STOLEN')),
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

-- 2. Stolen & Blacklisted Documents (Interpol / MHA Watchlist)
CREATE TABLE IF NOT EXISTS blacklist (
  passport_number TEXT PRIMARY KEY,
  reason TEXT NOT NULL,
  flagged_by TEXT DEFAULT 'Interpol SLTD',
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

-- 3. Cryptographic Screening Audit Ledger (Immutable Hash Chain)
CREATE TABLE IF NOT EXISTS screening_audit_ledger (
  block_index INTEGER PRIMARY KEY AUTOINCREMENT,
  timestamp TEXT NOT NULL,
  previous_hash TEXT NOT NULL,
  document_hash TEXT NOT NULL,
  risk_score REAL NOT NULL,
  risk_level TEXT NOT NULL,
  reasons_json TEXT NOT NULL,
  officer_id TEXT NOT NULL,
  model_versions TEXT NOT NULL,
  block_hash TEXT NOT NULL
);
```

8.2 Complete REST API Contract Specification

METHOD	ENDPOINT	PAYLOAD / PARAMETERS	RESPONSE SCHEMA
POST	/api/upload	multipart/form-data (file: binary, doc_type: str)	{"file_id": str, "status": "uploaded"}
POST	/api/extract/{file_id}	None	Extracted fields (name, number, dates, gender, confidences).
POST	/api/validate/{file_id}	None	Checksums (7-3-1, composite), date validity, blacklist status.
POST	/api/tamper-check/{file_id}	None	ELA score, ELA image URL, EXIF software, ViT forgery score.
POST	/api/face-verify/{file_id}	multipart/form-data (live_face: binary)	{"match": bool, "similarity": float, "faiss_alias_alert": bool}
POST	/api/analyze/{file_id}	Optional live face payload	Unified Dashboard Response: Scores, bands, reasons, ledger block.
GET	/api/ledger/audit	None	Array of all immutable audit blocks with hash integrity status.

9. Implementation Roadmap & Sprint Breakdown

The project execution plan is structured into 5 focused development sprints:

SPRINT	PILLAR & FOCUS	CONCRETE ENGINEERING DELIVERABLES	STATUS
Sprint 1	Core CV, OCR & Rules	OpenCV perspective transform, PassportEye MRZ extraction, ICAO 7-3-1 checksums, auto-corrector.	COMPLETED
Sprint 2	Digital Forgery Forensics	JPEG Q=90 ELA heatmap generator, EXIF software parser, Hugging Face ViT forgery classification.	COMPLETED
Sprint 3	Biometric Engine & FAISS	Portrait face auto-crop, HTML5 live webcam stream in Next.js, ArcFace 1:1 cosine match, FAISS 1:N index.	IN PROGRESS
Sprint 4	Ledger & National IDs	Aadhaar Verhoeff checksum, VIZ-to-MRZ discrepancy check, SHA-256 Merkle audit ledger endpoints.	UPCOMING
Sprint 5	Demo Suite & Hardening	10-case synthetic test suite generation, performance benchmark tuning (<2.5s), presentation polish.	UPCOMING

10. Risk Management & Contingency Matrix

IDENTIFIED RISK	SEVERITY	IMPACT DESCRIPTION	ENGINEERED CONTINGENCY / MITIGATION
Webcam Unavailable	Medium	Evaluation laptop or booth has no functioning camera.	Pre-packaged test images allow simulated traveler face injection with 1-click fallback.
ViT Inference Delay	Medium	GPU unavailable on local evaluation machine (CPU fallback > 3s).	Model execution runs asynchronously; ELA and metadata scores provide immediate partial preview.
Passport MRZ Creased	Low	Folds or scratches cause character misreads.	Scoped positional letter-to-digit auto-corrector recovers OCR errors; generic OCR acts as fallback.
Database Unreachable	Low	External internet severed at remote border outpost.	Autonomous local SQLite registry stores all blacklist and revocation records on the edge.

Sign-off & Engineering Readiness

This PRD represents the complete functional and technical specification for **Project Guardian**. All modules, mathematical algorithms, data contracts, and operational requirements have been validated against the Ministry of Home Affairs / Sashastra Seema Bal PS 26188 mandate.